

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

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FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.3: TRIGONOMETRY

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. It is your chance to show everything you have learned across this unit by answering the driving question fully and clearly. Read each section prompt carefully, then write your best response. Use diagrams where they help. Draw on all the evidence, experiments, and activities from every lesson — especially your clinometer investigation. A strong answer will connect back to the anchoring phenomenon: the water tank at your school or homestead. Your teacher will assess your work using the rubric at the bottom of this document.

Section 1: Restating the Problem — Why Can't We Just Measure It?

In your own words, explain why it is difficult or impossible to measure the height of an elevated water tank directly using a tape measure. What information CAN you collect from the ground without

Measuring the height of a water tank directly is impractical because the tank is elevated on a steel frame several metres above the ground, and climbing it — or stretching a tape measure vertically along its full height — is unsafe and often not allowed. However, from a safe position on the ground, you CAN measure two things accurately: (1) the horizontal distance from where you are standing to the base of the tank's support frame, and (2) the angle of elevation — the angle between your line of sight to the top of the tank and the horizontal ground. These two pieces of information, a known side and a known angle in a right-angled triangle, are all that trigonometry needs to calculate the unknown height precisely — without ever touching the tank.

climbing? Describe the two pieces of information that make the trigonometric method work.

Section 2: The Trigonometric Tools — SOH-CAH-TOA and the Right-Angled Triangle

Draw and label a right-angled triangle that represents the water-tank situation. Identify the angle of elevation (θ), the horizontal distance (adjacent side), and the height of the tank (opposite side). Then explain which trigonometric ratio — sine, cosine, or tangent — is the most useful here and why. Write the formula you would use to calculate the height, and show a worked example using an angle of 35° and a horizontal distance of 12 metres.

In the right-angled triangle formed by the water-tank situation: the horizontal ground forms the ADJACENT side, the vertical height of the tank forms the OPPOSITE side, and the imaginary line from your eye to the top of the tank is the HYPOTENUSE. The angle of elevation θ sits between the adjacent side and the hypotenuse at ground level.

The most useful ratio here is the TANGENT, because we know the adjacent side (horizontal distance) and we want to find the opposite side (height) — and tangent connects exactly those two sides:

$$\tan(\theta) = \text{opposite} / \text{adjacent} \rightarrow \text{height} = \text{adjacent} \times \tan(\theta)$$

Worked example:

$$\theta = 35^\circ, \text{ horizontal distance} = 12 \text{ m}$$

$$\text{height} = 12 \times \tan(35^\circ)$$

$$\text{height} = 12 \times 0.7002$$

$$\text{height} \approx 8.40 \text{ metres}$$

So the top of the tank is approximately 8.40 metres above the ground. Note: if your eye level is above the ground (e.g., 1.5 m), you must add that value to your final answer to get the true height from the ground.

Section 3: The Consistency Check — Two Students, Two Distances, Same Height

Recall that in the anchoring phenomenon, TWO students stood at DIFFERENT distances from the water tank and measured DIFFERENT angles of elevation — but both should calculate the SAME height. Explain mathematically why this is true. Use a second worked example (angle = 52° , horizontal distance = 7.5 m) and show that both results agree. What does this consistency tell us about the reliability of trigonometry?

Because the water tank has one fixed height, any right-angled triangle drawn from a different position on the ground will have a different base length and a different angle — but the vertical side (the height) stays constant. Trigonometry captures this relationship precisely: as you move farther away, the angle gets smaller, and the larger base length compensates exactly to produce the same height.

Second worked example:

$$\theta = 52^\circ, \text{ horizontal distance} = 7.5 \text{ m}$$

$$\text{height} = 7.5 \times \tan(52^\circ)$$

$$\text{height} = 7.5 \times 1.2799$$

$$\text{height} \approx 9.60 \text{ metres}$$

If both students' results are close (differences only due to small measurement errors in reading the angle), this confirms that the method is internally consistent and reliable. This consistency is powerful: it means trigonometry is not a lucky guess — it is a dependable mathematical relationship. Kenyan surveyors use exactly this consistency check when mapping land in the Rift Valley;

multiple readings from different stations must agree before a boundary is accepted as legal.

Section 4: Complementary Angles and the Sine/Cosine Connection

During your lessons you discovered a shortcut involving complementary angles. Explain what complementary angles are, and describe the relationship between $\sin(\theta)$ and $\cos(90^\circ - \theta)$. How does this relationship help when solving trigonometry problems involving right-angled triangles? Give one example using the water-tank triangle to illustrate this connection.

Two angles are COMPLEMENTARY when they add up to 90° . In any right-angled triangle, the two non-right angles are always complementary — if one angle is θ , the other is $(90^\circ - \theta)$.

The shortcut is: $\sin(\theta) = \cos(90^\circ - \theta)$, and $\cos(\theta) = \sin(90^\circ - \theta)$. This means the sine of an angle equals the cosine of its complement, and vice versa.

Water-tank example:

In the triangle with $\theta = 35^\circ$ (angle of elevation), the angle at the top of the tank is $90^\circ - 35^\circ = 55^\circ$.

$$\sin(35^\circ) = \cos(55^\circ) \approx 0.5736$$

$$\cos(35^\circ) = \sin(55^\circ) \approx 0.8192$$

This is useful because if a problem gives you the angle at the top of the triangle instead of the base, you do not need to re-draw the triangle — you can simply swap sine and cosine using the complementary relationship. It also helps you check your calculator work: if $\sin(40^\circ) \neq \cos(50^\circ)$ on your calculator, you have made an error.

Section 5: Answering the Driving Question — Your Full Explanation

Now bring everything together. Answer the driving question fully: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly?' Reference the water tank, your clinometer investigation, and at least ONE real-world Kenyan application beyond the school compound. Explain why this skill matters.

We can find any unknown height or distance — without measuring it directly — by forming an imaginary right-angled triangle between ourselves, the object, and the ground, then applying a trigonometric ratio to the known side and known angle.

For the water tank: stand at a measured horizontal distance from the base, use a clinometer (like the one we built from a protractor and string) to measure the angle of elevation to the top of the tank, then calculate: height = horizontal distance $\times$ $\tan$ (angle of elevation). The height is revealed precisely, with no climbing required.

Our clinometer investigation confirmed this works in practice. When two of us measured from different distances, our calculated heights differed by less than 0.3 metres — a small error caused only by the precision of our protractor reading, not a flaw in the method.

Beyond the school compound, this same approach is used every day across Kenya:

- Foresters in the Mau Forest measure the height of indigenous trees to estimate timber volume — without felling a single tree.
- Engineers designing bridges along the Thika Superhighway calculate vertical clearance heights using angle and distance measurements taken from the road.
- Tea farmers in Kericho use slope angles to plan drainage channels, preventing soil erosion on hillside farms.

Trigonometry matters because it gives us a universal mathematical tool for bridging what we CAN measure (a safe, horizontal distance and a simple angle) to what we CANNOT measure directly (a dangerous or inaccessible height). It turns an impossible

	physical task into a straightforward calculation — and that is exactly why it is one of the most practical branches of mathematics we will ever study.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of the Right-Angled Triangle Model	Correctly identifies and labels all three sides (opposite, adjacent, hypotenuse) and the angle of elevation in the context of the water-tank scenario. Diagram is accurate and clearly annotated.	Correctly identifies most sides and the angle of elevation. Minor labelling error that does not affect the calculation.	Shows a triangle but sides are mislabelled or the angle of elevation is placed incorrectly. Diagram is incomplete or missing.
Selection and Application of Trigonometric Ratios (SOH-CAH-TOA)	Selects the correct ratio (tangent) with a clear, explicit justification. Applies the formula accurately in at least two worked examples, showing all steps and arriving at correct answers.	Selects the correct ratio and applies it correctly in at least one worked example. Steps are mostly shown but one minor arithmetic or rounding error is present.	Attempts to use a trigonometric ratio but selects the wrong one, or applies the formula incorrectly. Working is incomplete or the answer is not obtained.
Explanation of the Consistency Check (Two Angles, Same Height)	Clearly explains why two different angles from two different distances must yield the same height. Uses a second worked example to verify, and connects the idea to the reliability and real-world use of trigonometry (e.g., surveying in Kenya).	Explains the consistency idea and attempts a second worked example. The conceptual explanation is correct but the real-world connection is vague or missing.	Acknowledges that two students get different angles but cannot explain why the height should be the same. No second worked example attempted.
Complementary Angles and the Sin/Cos Relationship	Accurately defines complementary angles and states the identity $\sin(\theta) = \cos(90^\circ - \theta)$ with a correct example drawn from the water-tank triangle. Explains how this identity is practically useful for checking work or solving alternative problem setups.	Correctly states the complementary angle identity and gives a valid example. Explanation of practical usefulness is present but brief.	Attempts to discuss complementary angles but the identity is stated incorrectly or the example contains errors. Practical usefulness is not explained.
Quality of the Final Driving-Question Response	Provides a full, coherent answer to the driving question that integrates the water-tank phenomenon, clinometer investigation findings, and at least one specific Kenyan real-world application. Explains why the skill is important in clear, student-accessible language.	Answers the driving question and references the water-tank phenomenon and one real-world application. The clinometer investigation is mentioned but not discussed in detail. Explanation is mostly clear.	Attempts to answer the driving question but the response is incomplete, missing the phenomenon connection or real-world application. Explanation is unclear or uses incorrect terminology throughout.